Building quantum codes with transversal CCZ

from projective geometry

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Joint work with Kenneth R. Brown

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arXiv:2610.09341 Code and data

A code can protect information and implement a gate

The construction

48 physical qubits encode 3 logical qubits with distance 3. A layer of $26\,T + 22\,T^\dagger$ implements logical CCZ.

- A *logical qubit* is one qubit encoded across a larger block.
- Distance 3 permits correction of an arbitrary single-qubit error.
- A *transversal* layer acts separately on each physical qubit.

The code must satisfy both error-correction constraints and gate constraints.

Binary codes describe a quantum encoding

Let S have independent stabilizer rows and K have k logical rows.

$$\mathcal{C}_2 = \text{rowspan}(S) \subset \mathcal{C}_1 = \text{rowspan} \begin{bmatrix} K \\ S \end{bmatrix}.$$

A CSS code has the orthonormal logical basis

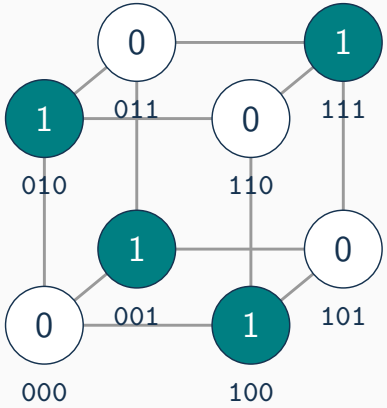
$$|x\rangle_L = \frac{1}{\sqrt{|\mathcal{C}_2|}} \sum_{v \in \mathcal{C}_2} |xK \oplus v\rangle, \qquad x \in \mathbb{F}_2^k.$$

Rows of K are independent modulo $\mathcal{C}_2$.

X flips a physical bit. Z changes its sign when that bit is 1. The X checks come from $\mathcal{C}_2$; the Z checks come from $\mathcal{C}_1^\perp$.

Logical basis states are equal superpositions over distinct classical cosets.

The eight-qubit cube starts with affine functions



Physical positions: $u \in \mathbb{F}_2^3$.

$$f_{x,b}(u) = b \oplus x_1u_1 \oplus x_2u_2 \oplus x_3u_3.$$

The truth tables form $\mathrm{RM}(1,3)$, an $[8,4,4]$ binary code.

Take $S = 1$ and $K = (u_1, u_2, u_3)^\top$.

The same cube supplies physical positions, generator rows, and a gate pattern.

Complementary words encode three logical qubits

The classical code has 16 words, partitioned into 8 complementary pairs:

$$|x\rangle_L = \frac{|f_{x,0}\rangle + |f_{x,1}\rangle}{\sqrt{2}}, \qquad x \in \mathbb{F}_2^3.$$

For example,

$$|000\rangle_L = \frac{|00000000\rangle + |11111111\rangle}{\sqrt{2}}.$$

The eight disjoint pairs give eight orthogonal states: $k = 3$ logical qubits.

Flipping all 8 bits exchanges the two words and leaves the encoded state unchanged.

A diagonal gate must assign the same phase to both words of each pair.

Single-qubit phases can produce a three-qubit CCZ

$$T = \mathrm{diag}(1, e^{i\pi/4}), \qquad T^\dagger = T^{-1}.$$

Logical CCZ multiplies $|111\rangle_L$ by -1 and leaves the other 7 logical labels unchanged:

$$\mathrm{CCZ}|x\rangle_L = (-1)^{x_1 x_2 x_3}|x\rangle_L.$$

Choose $T^\dagger$ at parity-one vertices $p(u) = u_1 \oplus u_2 \oplus u_3$ and T elsewhere.

$$W(c) = \sum_u (-1)^{p(u)} c(u), \qquad \text{phase on } |c\rangle = e^{i\pi W(c)/4}.$$

The toy gate works, but its distance is two

Since $W(c) = d_H(p, c) - 4$,

logical label	$W(f_{x,0}),\ W(f_{x,1})$	phase
$x \neq 111$	$0,\ 0$	$+1$
$x = 111$	$-4,\ +4$	-1

Both words in each pair acquire the required phase.

Two physical Z errors commute with the all-X check and can change a logical sign. Thus the cube code has parameters $[\![8, 3, 2]\!]$.

The 48-qubit construction keeps the phase mechanism and raises the distance to 3.

Overlap counts control weights of sums

For binary words a, b, their product $a \wedge b$ selects the common ones.

$$\mathrm{wt}(a \oplus b) = \mathrm{wt}(a) + \mathrm{wt}(b) - 2\,\mathrm{wt}(a \wedge b).$$

For three words the inclusion–exclusion coefficients are $1, -2, 4$:

$$\begin{aligned} W(a \oplus b \oplus c) = {} & W(a) + W(b) + W(c) \\ & - 2\big(W(a \wedge b) + W(a \wedge c) + W(b \wedge c)\big) \\ & + 4W(a \wedge b \wedge c). \end{aligned}$$

The identity holds for ordinary or signed weights. Terms of order 4 vanish modulo 8.

The general gate condition is a signed-weight congruence

Write $\Gamma_j = +1$ for T and -1 for $T^\dagger$; define

$$W_\Gamma(c) = \sum_{j=1}^{n} \Gamma_j c_j.$$

A physical gate layer implements logical CCZ exactly when

$$W_\Gamma(xK \oplus yS) \equiv 4x_1x_2x_3 \pmod 8$$

for every $x \in \mathbb{F}_2^3$ and $y \in \mathbb{F}_2^s$.

Dependence on x produces the logical gate. Independence from y preserves each stabilizer coset.

A gate-design question becomes an integer congruence on a binary code.

Only single, pair, and triple overlaps are needed

For distinct generator rows g_a, g_b, g_c of $G = [K; S]$:

$$W_\Gamma(g_a) \equiv 0 \pmod 8,$$
$$W_\Gamma(g_a \wedge g_b) \equiv 0 \pmod 4,$$
$$\mathrm{wt}(g_a \wedge g_b \wedge g_c) \equiv \begin{cases} 1 \pmod 2, & \{g_a, g_b, g_c\} = \{K_1, K_2, K_3\}, \\ 0 \pmod 2, & \text{otherwise.} \end{cases}$$

The parity conditions alone allow a diagonal Clifford correction: *quasi-transversal* CCZ. The signed congruences demand physical $T/T^\dagger$ alone.

All nine parity conditions

Geometry fixes the stabilizers before the search

1. Choose distinct nonzero columns for S: guarantee $d_Z \geq 3$.
2. Choose an 8-divisible stabilizer code: every stabilizer weight is a multiple of 8.
3. Search the logical rows K, then test physical signs Γ.

8-divisibility enforces the stabilizer-only overlap parities. The mixed and logical-row conditions remain to be solved.

The solver does not rediscover distance or the geometric stabilizer structure.

Projectivity rules out weight-one and weight-two Z logicals

Let β_j be column j of S. A Z error at site j has syndrome β_j.

$$\beta_j \neq 0 \quad \Longrightarrow \quad \text{every weight-one error is detected,}$$

$$\beta_i \neq \beta_j \quad \Longrightarrow \quad \beta_i \oplus \beta_j \neq 0.$$

Thus distinct nonzero columns detect every weight-two Z error.

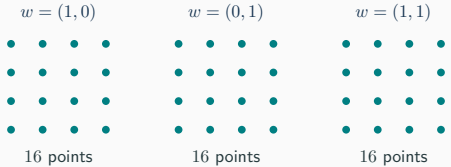
Over $\mathbb{F}_2$, those columns are points of $\mathrm{PG}(s-1, 2)$.

$d_Z \geq 3$ follows from the stabilizer support, independently of the logical labels.

The support is projective space minus a subspace

$$\mathcal{P} = \mathrm{PG}(5,2) \setminus \mathrm{PG}(3,2), \qquad |\mathcal{P}| = 63 - 15 = 48.$$

Write $\beta = (u, w) \in \mathbb{F}_2^4 \times \mathbb{F}_2^2$ and retain exactly $w \neq 00$.



The 6 coordinate functions $u_1, \ldots, u_4, w_1, w_2$ generate S.

Three cosets, each with 16 positions, give a projective $[48, 6]$ stabilizer code.

Hyperplane sections give the two nonzero weights

A stabilizer word is the evaluation of

$$\ell(u, w) = \lambda \cdot u \oplus \mu \cdot w.$$

- $\lambda \neq 0$: balanced on each of 3 blocks, weight $3 \cdot 8 = 24$.
- $\lambda = 0$, $\mu \neq 0$: one on 2 of the 3 blocks, weight $2 \cdot 16 = 32$.

$$\mathrm{wt}(\ell) = |\mathcal{P}| - |\mathcal{P} \cap \{\ell = 0\}|.$$

Weights 24 and 32 are multiples of 8. Geometry supplies the divisibility.

Calderbank–Kantor: two-weight codes and finite geometry.

SAT assigns three logical bits to each geometric point

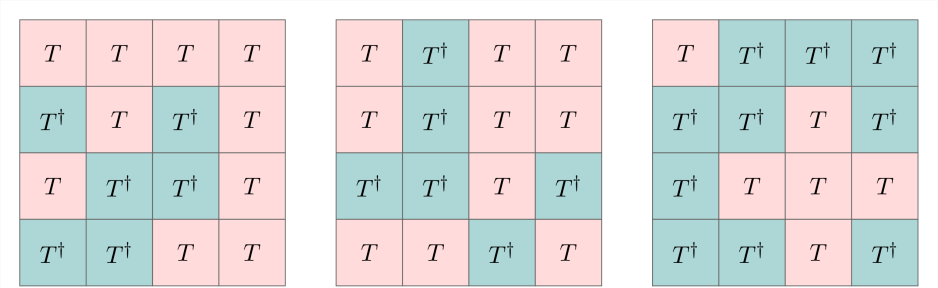
Unknowns: the column labels $\alpha_j \in \mathbb{F}_2^3$ of K.

$$3n = 144 \text{ label bits}, \qquad 7 + 6s + 3\binom{s}{2} = 88 \text{ parity constraints.}$$

- Stabilizer-only conditions already hold.
- Distance constraints are absent because S guarantees $d_Z \geq 3$.
- Remaining parity conditions include products of up to three label bits.
- A compatible labeling is followed by a separate physical-sign feasibility test.

Explicit logical rows

The published pattern is a layer of 48 phase gates



$26\,T + 22\,T^\dagger \quad \Longrightarrow \quad$ logical CCZ on 3 encoded qubits.

Pink: T. Teal: $T^\dagger$. Three 16-qubit blocks.

Pattern formula

Lu and Brown, arXiv:2610.09341, Figure 1 (grid panels).

A three-point witness makes the Z-distance exact

Put a Z error at $u = 0000$ in each of the 3 blocks.

$$\underbrace{10 \oplus 01 \oplus 11}_{\text{stabilizer syndrome}} = 00, \qquad \underbrace{011 \oplus 110 \oplus 000}_{\text{logical syndrome}} = 101 \neq 000.$$

The operator commutes with every check but acts nontrivially on encoded information.

$$Q_{48} = [\![48, 3, 3]\!], \qquad (d_X, d_Z) = (16, 3).$$

Projectivity proves $d_Z \geq 3$; the explicit witness proves $d_Z \leq 3$.

The lower bound turns supports into Reed–Muller words

For a projective support $\mathcal{P} \subset \mathbb{F}_2^s$, let $\chi_{\mathcal{P}}$ be its Boolean indicator. The stabilizer-overlap parities constrain its algebraic degree.

$$c \in \mathrm{RM}(s-4, s), \qquad n = \mathrm{wt}(c) - c(0).$$

The indicator may differ from c at the origin.

A short code must therefore arise from a low-weight Reed–Muller word, followed by compatible logical labels.

Classification of low-weight Boolean polynomials supplies a finite exclusion problem.

Indicator correspondence

Our exclusion argument covers every length through 38

1. Reduce stabilizer rank to $s \leq 8$ when $n \leq 38$.
2. Exclude $s \leq 5$, then settle ranks 6 and 7 using the weight spectrum and support geometry.
3. At rank 8, reduce to finitely many normal forms and enumerate the remaining cases.
4. Show that no remaining support admits the required logical triple.

Independent result of our paper

No CSS code with $k = 3$, $d_Z \geq 3$, and the nine CH conditions has $n \leq 38$.

Lu and Brown, Theorem 4.5 and Appendix D.4; cited Reed–Muller classifications are inputs.

Concurrent classification closes the 39–46 gap

Length	Status	Evidence
$n \leq 38$	Excluded	Our lower bound
$39 \leq n \leq 46$	Excluded	Wills–Jain–Singh
$n = 47$	Quasi-transversal CCZ	Jacinto et al.
$n = 48$	Physical $T/T^\dagger$ only	Our explicit code

Wills–Jain–Singh classify $n \leq 54$, $d_Z \geq 3$, up to CNOT+S output equivalence. The nine CH conditions select their CCZ output class.

Their classification does not test physical-sign feasibility for every code.

What is proved at 47?

Wills, Jain, Singh: 2609.30860 Jacinto et al.: 2606.07734

The geometric construction produces a family

Choose a subspace complement

$$\mathrm{PG}(p+q-1,2) \setminus \mathrm{PG}(p-1,2), \qquad n = 2^{p+q} - 2^p = 2^p(2^q - 1).$$

The paper constructs codes at 13 block lengths from 48 to 496, with $d_Z = 3$ or 4.

Each support requires compatible logical rows and physical signs. Existence is not automatic from the point count.

The next algebraic task is to characterize which supports admit the required completions.

Lu and Brown, Corollary 2.4 and Table 3.

Gate-aware search complements general SAT code design

DalFavero et al.

- Discover or extend stabilizer codes.
- Prescribe detectable or correctable errors.
- Incorporate symmetries and hardware noise.

This construction

- Fix geometric stabilizers first.
- Search the logical rows for CCZ.
- Test the physical gate signs.

Error-detection constraints alone do not guarantee fault-tolerant logical operations.

DalFavero et al., arXiv:2608.23460, Sections VI and Appendix E.

Open questions connect algebra to implementation

- Characterize geometric supports admitting compatible logical rows and physical signs.
- Construct higher-distance families and extensions to other non-Clifford gates or more logical qubits.
- Determine the minimum length for physical $T/T^\dagger$ CCZ without a correction.
- Compare full circuits, noise, and resource costs before claiming a practical advantage.

Geometry supplies distance and divisibility; logical labels and signed weights supply the gate.

Joint work with Kenneth R. Brown. Thanks to Robert Calderbank for classical coding-theory pointers.

Paper: 2610.09341 Code and data

The nine parity conditions

K1	$\mathrm{wt}(K_i) = 0 \pmod 2$
K2	$\mathrm{wt}(K_i \wedge K_j) = 0 \pmod 2$
K3	$\mathrm{wt}(K_1 \wedge K_2 \wedge K_3) = 1 \pmod 2$
KS1	$\mathrm{wt}(K_i \wedge S_l) = 0 \pmod 2$
KS2	$\mathrm{wt}(K_i \wedge K_j \wedge S_l) = 0 \pmod 2$
KS3	$\mathrm{wt}(K_i \wedge S_l \wedge S_m) = 0 \pmod 2$
S1	$\mathrm{wt}(S_l) = 0 \pmod 2$
S2	$\mathrm{wt}(S_l \wedge S_m) = 0 \pmod 2$
S3	$\mathrm{wt}(S_l \wedge S_m \wedge S_r) = 0 \pmod 2$

Distinct logical indices $i < j$; stabilizer indices $l < m < r$.

K1–K3 select CCZ: $\tau_{iii} = \tau_{iij} = 0$ and $\tau_{123} = 1$.

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A logical row is a function on each block

The table specifies K_1:

w	10	01	11
$K_1(u, w)$	$u_1 \oplus u_3$	$1 \oplus u_3$	u_4

K_2 and K_3 are quadratic on each block (paper, Table 2). Evaluating the functions at all 16 addresses gives each block of the row.

The three rows must be independent modulo $\mathcal{C}_2$ and satisfy the overlap parities. A candidate labeling still needs a feasible physical-sign pattern.

Full matrices and independent verification are in the public repository.

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The physical signs have a polynomial formula

Set $\Gamma_j = 1 - 2\gamma_j$, where $\gamma_j \in \{0, 1\}$ marks $T^\dagger$.

$$\gamma \cdot c \equiv \frac{\mathrm{wt}(c)}{2} - 2x_1x_2x_3 \pmod 4, \qquad c = xK \oplus yS.$$

The unknowns remain binary even though the congruence is modulo 4.

$$\begin{aligned} \gamma(u, w) &= u_3 \oplus w_1 Q_1(u) \oplus w_2 Q_2(u), \\ Q_1 &= u_1 \oplus u_2 \oplus u_3 \oplus u_1u_3 \oplus u_2u_3 \oplus u_2u_4, \\ Q_2 &= u_3 \oplus u_4 \oplus u_1u_2 \oplus u_1u_4 \oplus u_3u_4. \end{aligned}$$

The resulting 22 flips satisfy the phase condition on all $8 \times 64 = 512$ codewords.

Lu and Brown, Proposition 3.9.

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Why a Reed–Muller indicator appears

Over $\mathbb{F}_2$, the stabilizer conditions say

$$\sum_{\beta \in \mathcal{P}} \beta_{i_1} \cdots \beta_{i_t} = 0, \qquad 1 \le t \le 3.$$

Let $c = \chi_{\mathcal{P}} + \epsilon \delta_0$, choosing ϵ so $\mathrm{wt}(c)$ is even. The origin changes no positive-degree moment. Then $c \perp \mathrm{RM}(3, s)$, hence

$$c \in \mathrm{RM}(3, s)^{\perp} = \mathrm{RM}(s - 4, s), \qquad n = \mathrm{wt}(c) - c(0).$$

Rank reduction and cited low-weight classifications make the $n \le 38$ search finite.

Lu and Brown, Theorem 4.1 and Appendix D.

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What the 47-qubit comparison establishes

Jacinto et al. give a $[\![47, 3, 3]\!]$ code satisfying the nine CH conditions. Its physical $T/T^\dagger$ sign system is infeasible (paper, Example 3.7).

Wills–Jain–Singh establish minimum length 47 in the quasi-transversal CCZ class with $k = 3$, $d_Z \geq 3$. Their catalogue identifies outputs up to CNOT and S, not full Clifford equivalence.

These facts do not establish uniqueness of the length-47 code or rule out every other length-47 completion with feasible signs.

Jacinto et al.: 2606.07734. Wills–Jain–Singh: 2609.30860.

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Code length and distillation footprint are different resources

Apply the phase gates to encoded $|+\rangle^{\otimes 3}$, measure checks, and decode to prepare a CCZ resource state.

- n: code length, or number of noisy T resource inputs in the distillation description.
- N: circuit footprint in the cited distillation implementation; for the matrix description, $N = k + s$.
- Jacinto's 47-to-1 distance-3 CCZ protocol has $(n, N) = (47, 9)$.

At least 3 input Z errors are needed for an undetected logical error under the ideal-Clifford distillation model. A complete physical factory additionally needs a circuit, preparation, measurements, decoding, and a noise model.

Jacinto et al., 2606.07734; Wills–Jain–Singh, 2609.30860, Table 2.

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